Murgitroyd

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Murgitroyd House
165-169 Scotland St.
Glasgow
G5 8PL

info@murgitroyd.com
+44 [0] 141 307 8400

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80298 Munich
Germany

European Patent Application No 22904861.6
Western Digital Technologies, Inc.
IP Title: Devices, Systems, And Methods Of Using Smart Fluids To Control Molecule Speeds

In response to the Communication pursuant to Rule 70(2) and Rule 70a(2) EPC, dated 25th November 2025, the Applicant confirms that he does wish to proceed with examination of the above-referenced European patent application.

This letter is also in response to the objections raised in the Search Opinion accompanying the Report dated 5th November 2025. We file herewith:

- an amended claim set
- a marked-up copy of the claim amendments
- amended description pages 1, 2, 2a and 30
- a marked-up copy of the amended description pages

Amendments
Claims 1 to 15 have been amended to insert reference signs throughout (addressing the objection at item 4.1).

The main claims have not been cast in two-part form since it is considered that to do so would reduce the clarity of the claims and offend against Article 84 EPC [item 4.2 noted but disputed].

In the specification, amendments have been made on pages 2 and 2a to bring the consistory clauses into conformity with the amended claims (see item 4.4) and to acknowledge prior art documents D1 and D2 (see item 4.3). Page 1 is amended to remove an instance of incorporation by reference. Page 30 is amended to remove mention of the "spirit" of the invention.

For the avoidance of doubt, the Applicant notes that where subject matter is deleted in the course of the above amendments, this deletion is made without prejudice and without abandonment of subject matter. In particular, the Applicant reserves the right to make any or all of the excised subject matter the basis for one or more divisional applications.

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Added Subject Matter

In item 2 of the Written Opinion, the Examiner requests detailed basis for previously filed amendments in claims 3, 4, 8, 14 and 15. The Examiner is invited to refer to the marginal notes in the annotated version of the newly submitted claim set.

The present application addresses solving a key challenge in nanopore sequencing: the high speed at which molecules, such as DNA, pass through a nanopore. This high speed makes it difficult to read the molecular sequence with high resolution and a good signal-to-noise ratio. The core of the invention is a speed-control device that works in conjunction with the nanopore. This device creates mechanical resistance (i.e. viscosity in an electrorheological or magnetorheological fluid) to slow down the molecule's movement.

Here, it will be noted that claim 3 being dependent upon claim 2 (rather than claim 1) has support in the text through the text at paragraph [0150] since this paragraph explicitly links the embodiments with speed detector feedback (as in claim 2) to the embodiments with a field-response fluid that is **electrorheological** (ER) such as Figs 2A-2C where first and second electrodes are provided.

Likewise, paragraph [0150] also supports claim 4 being dependent upon claim 2 (rather than claim 1). It explicitly links the embodiments with speed detector feedback (as in claim 2) to the embodiments with a field-response fluid that is **magnetorheological** (MR) or a **ferrofluid**, such as Figs 2D – 2G.

In relation to current claim 8, it is noted that the features of this claim are clearly supported in combination with any preceding system claim by paragraph [0153]. This paragraph unequivocally provides that preceding embodiments also apply to non-biological/inorganic molecules.

The extended dependency in claim 14 to claim 13 (rather than only claim 12) has basis in para [0160] since this paragraph contemplates a system having a combination of “means of directing” with “means for detecting the speed”. The next paragraph, para [0161], corresponds to the subject matter of claim 15 and refers back to "systems" - clearly implying the systems of the preceding paragraph.

Novelty - Articles 52(1), 54(1) & (2) EPC

Patentability - Claim 1

At items 3.1 & 3.1.1 of the EPO Communication, the Examiner argues that claims 1 and 12 respectively are anticipated by document D1 (WO2019213437A1).

The Examiner asserts that the subject-matter of claim 1 is anticipated by D1 based on paragraph [0160] of D1. Claim 1 recites:

> A system (100) for controlling movement of at least one molecule (20) in a first fluid having a first viscosity, the system (100) comprising:
>
> a fluid region (105) defined by at least one fluid-retaining surface (102), wherein the fluid region (105) is traversable by the at least one molecule

(20);

a field-responsive fluid (106) situated in the fluid region (105), wherein the field-responsive fluid (106) is preferably coupled to at least a portion of the at least one fluid-retaining surface (102); and

a field generator (110) for generating a magnetic or electric field (111) across the fluid region (105), wherein the field generator (110) preferably comprises a voltage source (121),

wherein, in response to a magnitude of the magnetic or electric field (111) across the fluid region (105) exceeding a threshold magnitude, a viscosity of the field-responsive fluid (106) is greater than the first viscosity.

The Examiner maps these features onto a single sentence fragment in paragraph [0160] of D1, which reads:

"slowing down the speed of the polymer, from ca. 1MHz to ca. 100-200Hz, for example using a medium comprising an electrorheological fluid in which becomes more viscous when a voltage is applied, thereby slowing down the speed of the polymer through the nanopore."

This passage does not directly and unambiguously disclose all features of claim 1, for the following reasons.

No disclosure of "a fluid region defined by at least one fluid-retaining surface"

Claim 1 requires a specific structural arrangement: a fluid region defined by at least one fluid-retaining surface, where the field-responsive fluid is situated in that fluid region. The fluid-retaining surface is a structural element that holds or retains the field-responsive fluid within a defined region. This is a specific physical structure through which the molecule traverses. Numerous figures in the application illustrate the fluid region in relation to the field-responsive fluid. *See, e.g.* , FIGS. 2B-2F, 3A-3E, and 4A and the discussions thereof.

Paragraph [0160] of D1 merely lists the concept of "using a medium comprising an electrorheological fluid" as one of numerous bullet points in a long list of possible approaches to slow DNA translocation speed. D1 does not describe any structure, device, or system that implements this concept. There is no disclosure of a fluid-retaining surface, a defined fluid region, or any structural arrangement for holding an electrorheological fluid in a pathway of a molecule.

D1's paragraph [0160] is part of a general discussion of possible approaches to address translocation speed. It reads as a wish list or speculative enumeration. The mere

mention of "using a medium comprising an electrorheological fluid" does not constitute a direct and unambiguous disclosure of a system having a fluid region defined by at least one fluid-retaining surface with a field-responsive fluid situated therein. A passing mention of a material without any structural context does not amount to an enabling disclosure of a system comprising defined structural elements.

No disclosure of "a field generator for generating a magnetic or electric field across the fluid region"

The Examiner states that the means for generating the voltage are "directly and unambiguously disclosed" by D1. The Applicant disagrees. What D1 states is simply that an electrorheological fluid "becomes more viscous when a voltage is applied." This is a statement of a material property, not a disclosure of a field generator as a component of a system. D1 does not disclose any specific component that generates the field across a defined fluid region. The nanopore-associated electrodes described throughout D1 (e.g., the sensing electrodes or steering electrodes) serve entirely different purposes (sequencing, steering the DNA), and D1 does not teach or suggest using a separate or dedicated field generator to apply a field across a fluid region containing a field-responsive fluid.

The Examiner appears to reason that because the electrorheological fluid requires a voltage to change viscosity, the means to apply that voltage must be implicitly disclosed. But this reasoning conflates the disclosure of a material property with the disclosure of a system component. Claim 1 requires a field generator as a structural element of the system, which D1 does not disclose.

D1 does not disclose the specific condition "a viscosity of the field-responsive fluid is greater than the first viscosity"

Claim 1 is directed to a system for controlling movement of at least one molecule in a first fluid having a first viscosity, where the viscosity of the field-responsive fluid is greater than that first viscosity when the field across the fluid region exceeds a threshold. This implies a system where the molecule is initially in a first fluid (e.g., an electrolyte solution) and encounters the field-responsive fluid as a separate region with higher viscosity. *See, e.g.* , FIGS. 2B-2F, 3A-3E, 4A and descriptions thereof (indicating that field-responsive fluid 106 is not the same as the surrounding fluid).

D1's paragraph [0160] does not describe this arrangement. At most, D1 contemplates using an electrorheological fluid as the medium through which the polymer passes, not as a distinct fluid region with a viscosity greater than a separate first fluid in which the molecule initially resides.

In summary, paragraph [0160] of D1 is a bare, non-enabling mention of a concept. It does not disclose the structural features recited in claim 1. Specifically, D1 does not disclose a fluid region defined by at least one fluid-retaining surface, a field generator for generating a field across that fluid region, or the specific relationship between the first viscosity and the viscosity of the field-responsive fluid. Claim 1 is novel over D1.

Claim 12

The Examiner asserts that claim 12 is not new over either D1 or D2 (Marziali, US2013/344498 A1). Claim 12 recites:

> A system (100) for controlling a speed of at least one molecule (20), the system (100) comprising:
>
> a field-responsive fluid (106), wherein a property of the field-responsive fluid (106) is responsive to an applied magnetic or electric field (111), wherein the property of the field-responsive fluid (106) is preferably a viscosity of the field-responsive fluid (106);
>
> means (102, 125A, 125B, 132, 133) for holding the field-responsive fluid (106) in a pathway of the at least one molecule (20); and
>
> means (110, 125A, 125B, 131, 132, 133) for generating the applied magnetic or electric field (111) across the field-responsive fluid (106).

Novelty over D1 (item 3.1.1)

With respect to D1, the same arguments presented above for claim 1 apply to claim 12. For the same reasons as discussed above regarding the "fluid region defined by at least one fluid-retaining surface," D1 does not directly and unambiguously disclose, at paragraph [0160] or elsewhere, "means for holding the field-responsive fluid in a pathway of the at least one molecule." Likewise, for the same reasons as discussed above regarding the "field generator for generating a magnetic or electric field across the fluid region," D1 does not directly and unambiguously disclose, at paragraph [0160] or elsewhere, "means for generating the applied magnetic or electric field across the field-responsive fluid."

Novelty over D2 (item 3.2)

The Examiner specifically cites paragraph [0291] and figures 10A-10D of D2 as allegedly disclosing "wherein a property of the field-responsive fluid is responsive to an applied magnetic or electric field, wherein the property of the field-responsive fluid is preferably a viscosity of the field-responsive fluid."

Paragraph [0291] of D2 states:

> "In other embodiments, the medium comprises a magneto-rheological fluid so that the medium has a viscosity that inherently varies with the applied magnetic field."

This is a single sentence in Example 13, which is entitled "Magnetic Control of Particle Mobility." Examining the full context of Example 13 (paragraphs [0288]-[0291]), the example describes several different ways of using magnetic fields to alter particle mobility. Paragraph [0291] is merely one of several enumerated possibilities, and it is a bare statement of concept without structural detail.

Critically, D2's Example 13 and figures 10A-10D describe systems for concentrating and separating particles (specifically nucleic acids) using SCODA methods, not systems for controlling the speed of molecule translocation through a defined fluid pathway. The context of D2 is fundamentally different from claim 12.

D2 does not disclose, at paragraph [0291] or elsewhere, "means for holding the field-responsive fluid in a pathway of the at least one molecule" or "means for generating the applied magnetic or electric field across the field-responsive fluid." First, D2 does not describe holding a magneto-rheological fluid in a defined pathway through which a molecule translocates. D2's figures 10A-10D show a medium in which particles move under the influence of fields, not a defined pathway with a held fluid. Second, although D2 mentions applying a magnetic field, the purpose is to alter mobility of particles within a medium for separation/concentration, not to generate a field across a defined fluid region to control translocation speed.

The Applicant submits that D2's disclosure of using a magneto-rheological fluid as a general medium for SCODA-based particle concentration does not directly and unambiguously disclose a system with means for holding a field-responsive fluid in a pathway of a molecule and means for generating a field across that fluid to control molecule speed.

In summary, the Applicant respectfully submits that claim 12 is novel over both D1 and D2.

Claim 9: Inventiveness over D1 (item 3.3)

The Examiner asserts that claim 9 is not inventive over D1. Claim 9 recites:

> A method of controlling movement of at least one molecule (20), the method comprising:
>
> generating a magnetic or electric field (111) across a fluid region (105)

situated in an apparatus (100), the fluid region (105) containing a field-responsive fluid (106), wherein a viscosity of the field-responsive fluid (106) is dependent on a magnitude of the generated magnetic or electric field (111);

detecting a speed of the at least one molecule (20) through the fluid region (105); and

adjusting the magnitude of the magnetic or electric field (111) based at least in part on the detected speed of the at least one molecule (20) through the fluid region (105).

Closest Prior Art

The Applicant accepts D1 as the closest prior art for the purposes of this analysis.

Distinguishing Features

As acknowledged by the Examiner, claim 9 differs from D1 in:

1. Detecting a speed of the at least one molecule through the fluid region; and
2. Adjusting the magnitude of the magnetic or electric field based at least in part on the detected speed of the at least one molecule through the fluid region.

These features define a closed-loop feedback control system in which the speed of the molecule is actively detected and used to adjust the applied field in real time.

Technical Effect

The technical effect of these distinguishing features is the provision of active, real-time control over the translocation speed of a molecule through the fluid region, allowing precise and adaptive adjustment of the field-responsive fluid's viscosity to achieve a desired translocation speed. This enables single-nucleotide resolution when reading biomolecules through a nanopore, as explained in paragraph [0007] of the application as filed.

Objective Technical Problem

The objective technical problem, formulated without hindsight and derived from the technical effect above, can be stated as:

How to provide active, real-time control over the translocation speed of a molecule through a fluid region containing a field-responsive fluid.

Non-Obviousness

The Examiner asserts that "a skilled person - when reading that 'the speed of the

polymer, from ca. 1MHz to ca. 100-200Hz, for example using a medium comprising an electrorheological fluid in which becomes more viscous when a voltage is applied, thereby slowing down the speed of the polymer through the nanopore' - would *immediately recognise that speed detector coupled to the controller are implicit when measuring the above mentioned frequencies* ."

The Applicant respectfully but firmly disagrees with this reasoning, which constitutes an impermissible ex post facto analysis.

First, paragraph [0160] of D1 does not describe "measuring" anything. The passage states that the electrorheological fluid would slow the speed of the polymer "from ca. 1MHz to ca. 100-200Hz." This is a statement of the anticipated effect of using an electrorheological fluid. It describes the result, not a measurement process. There is no suggestion in D1 that a speed detector would be used to detect the actual translocation speed in real time, or that the detected speed would be fed back to adjust the applied voltage.

Second, the "frequencies" mentioned in paragraph [0160] of D1 (1MHz and 100-200Hz) refer to estimated translocation rates, not to detected signals from a speed detector. D1 is describing the general range of speeds at which a polymer would be expected to move with and without the use of an electrorheological fluid. This is a theoretical statement about the expected performance, not a disclosure of a feedback system.

Third, even if one were to assume for the sake of argument that D1 implicitly contemplates some form of speed detection, there is no teaching or suggestion in D1 of "adjusting the magnitude of the magnetic or electric field based at least in part on the detected speed of the at least one molecule through the fluid region." D1's electrorheological fluid concept, to the extent it is disclosed at all, appears to contemplate a static application of voltage to achieve a general reduction in speed. There is no suggestion of a dynamic, feedback-controlled system that adjusts the field in response to real-time speed measurements.

Fourth, the Examiner's conclusion that "the skilled person would regard it as a normal design procedure to combine all the features set out in claim 9" is a conclusory statement without adequate technical justification. The Examiner has not identified any document that teaches or suggests the combination of (a) a field-responsive fluid in a defined fluid region, (b) speed detection, and (c) feedback-based adjustment of the applied field. The mere fact that feedback control systems exist in other contexts does not render

obvious the application of feedback control to adjust the viscosity of a field-responsive fluid based on detected translocation speed.

The skilled person starting from D1 would have no motivation to implement a closed-loop feedback system as claimed, because D1 does not even disclose an implementable system for using an electrorheological fluid. D1 merely mentions the concept in passing. Without a concrete system to build upon, there is no basis for the skilled person to arrive at the sophisticated feedback-controlled system of claim 9.

For all of the reasons discussed above, claim 9 involves an inventive step over D1.

Claim 9: Inventiveness over D2 (item 3.4)

The Examiner presents an alternative inventive step analysis starting from D2 as closest prior art, with the distinguishing feature being that "a viscosity of the field-responsive fluid is greater than the first viscosity." The Examiner asserts that "[t]here is no specific technical effect in the originally filed description linked to the fact that a viscosity of the field-responsive fluid is greater than the first viscosity," and "in the absence of such an effect, the problem to be solved by the present invention may therefore be regarded as how to modify or adapt D2 to provide for an alternative field-responsive fluid."

The Applicant respectfully disagrees.

Technical Effect

The technical effect of the feature "a viscosity of the field-responsive fluid is greater than the first viscosity" is explicitly described throughout the application, including at paragraphs [0089], [0093], [0096], [0098], and [00123] of the published application WO 2023/107153. The technical effect is that the field-responsive fluid, when activated, provides mechanical resistance greater than the resistance of the surrounding fluid, thereby effectively slowing the molecule as it passes through the fluid region. Without this feature (i.e., without the viscosity of the field-responsive fluid exceeding that of the first fluid), there would be no slowing effect on the molecule.

This is not merely an "alternative" to D2's disclosure. The specific threshold relationship between the viscosity of the field-responsive fluid and the first viscosity is integral to the function of the claimed system. When the field is applied and the field-responsive fluid's viscosity exceeds the first viscosity, the molecule experiences increased resistance and is slowed. When the field is removed or reduced below the threshold, the molecule moves at its normal speed in the first fluid.

Objective Technical Problem

The objective technical problem, formulated without hindsight, is:

How to provide controllable resistance to molecule movement such that the molecule can be selectively slowed when passing through a defined region.

Non-Obviousness (item 3.5)

The Examiner asserts that paragraph [0160] of D1 discloses an electrorheological fluid "as providing the same advantages as in the present application" and that "the skilled person would therefore regard it as a normal option to include this feature in D2."

The Applicant respectfully submits that this reasoning is flawed for several reasons. First, D2 and D1 are directed to fundamentally different technical problems. D2 is directed to SCODA-based enrichment and concentration of nucleic acids. D1 is directed to synthesizing and reading data stored in polymers using nanopore devices. The skilled person working in the field of D2 (nucleic acid separation/concentration) would not look to D1 (DNA data storage) for solutions, as the technical contexts are entirely different.

Second, even if the skilled person were to consider D1, paragraph [0160] of D1 is a speculative, non-enabling mention of electrorheological fluids. As discussed above, D1 does not disclose an implementable system. The skilled person could not derive from D1 a teaching to modify D2 to include the specific feature that the viscosity of the field-responsive fluid exceeds the viscosity of the first fluid.

Third, D2's use of magneto-rheological fluids (paragraph [0291]) is in the context of altering particle mobility within a medium for SCODA concentration. The concept of having a separate fluid region with a viscosity greater than that in a surrounding first fluid is not taught or suggested by D2. D2 uses the magneto-rheological fluid as the overall medium, not as a separate region with specifically controlled higher viscosity relative to a first fluid.

Fourth, the combination of D2 and D1 requires the skilled person to (a) extract a single speculative sentence from D1, (b) interpret it as teaching that the viscosity should exceed that of a first fluid, and (c) apply this to D2's completely different system. This is a textbook example of impermissible hindsight reasoning, working backward from the claimed invention to construct a path through the prior art that would not have been apparent to the skilled person before the invention.

In summary, claim 9 involves an inventive step over D1 and D2, whether considered separately or in combination.

Dependent Claims (item 3.6)

The dependent claims are new and inventive for at least the same reasons that the independent claims from which they depend are new and inventive. Moreover, the

additional features of the dependent claims (e.g., specific speed detector and controller configurations, specific electrode arrangements, specific magnetorheological fluid arrangements with ferromagnetic yokes and solenoids, hollow cylinders, etc.) provide additional layers of specificity and technical contribution that further distinguish the claimed subject-matter from the cited prior art.

General Matters

We believe that all of the Examiner's objections have been addressed and we look forward to receiving a substantive examination report or notice of allowance in due course.

In the event that any matters remain outstanding, the Examiner is invited to contact the undersigned by telephone in order that those issues be speedily resolved.

As a formal precautionary measure, in the event that the Examining Division is minded to refuse this application without further written procedure, we hereby request oral proceedings.

Yours faithfully

DAVID KESTON
david.keston@murgitroyd.com

MURGITROYD & COMPANY
Professional Association No 834

EPPCT - RESPONSE RULE 70a(2)